1 Introduction

Each spring, in ecosystems across the temperate zone, ecological communities re-assemble form their period of winter dormancy with the hatching of insects, arrival of migratory birds, the germination of seeds and emergence of new leaves and flowers. The phenology—or seasonal timing—of these re-occurring life cycle events can be quite different, even among closely related organism in a shared habitat (). There is growing evidence that these phenological differences among taxa play a major role in dictating community interactions and shaping the structure and function of ecological communities ().

Phenological differences among species can generate seasonal priority effects; when earlier active species gain a competitive advantage by accesses limiting resources before their competitors emerge or arrive (). Recently the role of SPE's in coexistence has been demonstrated ()(Say better). Basically, coexistence between stronger and weaker competitors can occur when phenological difference function as a stabilizing mechanisms; (i.e., weaker competitor because active first, thus reducing the competitive advantages of the stronger intrinsic competitor). These dynamics indicate there is an ecological tradeoff between phenology and competitive ability in the context of coexistence.

Yet mechanistically characterizing this tradeoff is a challenge. This is because phenology is not a trait, but rather the realized product of an interaction between the environment (cues) and species unique physiology and genetic and developmental attributes. Environmental cues are changing rapidly, in fact phenological shifts are one of the most pronounced indicators of climate change to date (). However species' phenological responses to climate change differ significantly. We suggest that the degree to which a species responds to environmental variation known as **phenological sensitivity** (Δ phenology/ Δ environmental cue) is the primary mediator of the tradeoff between phenology and competition, and understanding this trait is key to predicting community assembly in an era of global change.

Below, we outline what is needed to better integrate phenological sensitivity into community assembly and develop a more robust way to predict assembly (say better and maybe tone down grandeur). First, we discuss how the the environment mediates phenological differences among species. Next, we briefly review how phenological is currently incorporated into assembly models. Then we implement a new germination module into an existing assembly model of Wolkovich and Donohue (can we say this), and run it under two climate scenario to understand how the tradeoff between phenology and competition is likely to shift. Finally, we outline a research agenda to better incorporating phenological dynamics into our understanding of community assembly that would substantially advance our ability to predict species interacts in an era global change.

2 How phenological differences arise

Environmental cues such as temperature, light and moisture are the primary environmental cues that control phenology (). Many species respond to multiple aspects of cues. For example some seeds respond to light levels and daylength (). Many species in temperate zone require exposure to cool winter temperature (0-4C, hereafter: winter chilling) to release dormancy in addition to exposure to warm spring temperatures to initiate phenological activity () Species are uniquely sensitive to these cues based on physiology, genetics, etc. For example, seeds have different kinds of dormancy. Plants and animals may relay on different cues. What this means that species individual phenology is response to environment, so the realized phenology differences with vary depending on the environmental conditions.

We illustrate this point by re-analyzing seed phenology germination data from ()(Buonaiuto & Wolkvich, see extended Methods for details). 11 co-occuring species were germinated under varying temperature conditions. We can see that even species from a common environment have very different sensitives to temperature cues (Figure 5a). As seen in our comparison of two species, *Persicaria virginiana* and *Eurybia divaricata*—a pair of woodland herbs that frequently compete in Eastern Temperate Forests ()—these sensitivities differences are manifest in changing realized phenological difference across environmental gradients (Figure 5b). For example, given the unique chilling sensitivites of the each species, under 6 weeks of chilling exposure *E. divaricata* would be predicted to germinate X days before *P. virginiana*, under 9 weeks of chilling the two species would be predicted to germinate roughly at the same time, and under 12 weeks of chill *P. virginana* would be predicted to germinate roughly Y days before *E. divaricata*. We can see here, that if we were to try and insert these two species into the classic phenology competition trade-off of seasonal priority effects (i.e early phenology/weaker competitor vs. late phenology) it could only work under one of these average climate scenarios. It follows that climate change, which shifts the mean environmental conditions species experience, would alter any stablized tradeoffs between priority effects and competition, and disrupt patterns of assembly. However, accounting for this (say better) this will require new approaches to modeling phenology in assembly models.

3 Recent work integrating phenology into assembly models

Growing interest in how phenology may structure communities has led to increased modeling efforts, with most approaches centering on the relationship between timing and competition (**???**). Superficially, phenological differences may seem to promote coexistence through niche differences—if species can perfectly partition a growing season such that they never compete together. In reality, however, limited resources often confer a fitness advantage to early-species via priority access to resources or otherwise create fitness hierarchies amongst species. Thus, adding phenological variation to species in coexistence models alone is generally destabilizing and requires additional variation that effectively reduces the competition between species to stabilize such communities (**?**). Models have done this through an explicit connection of phenological differences to parameters controlling the strength of competition (**?**), or implicitly by providing a fitness advantage to earlier species that then must be offset in communities where species also vary in parameters related to R^* (**??**).

These models have all recognized the importance of environmental variability in communities structured in part by phenological differences, generally including interannual variation in the environment that in turn drives differences in phenology. Many models—focused often on grasslands and water competition—have assigned phenology as equivalent to an intrinsic species-level timing (or season length strategy). Year to year variation in the environment then alters absolute differences between species, which then consistently co-varies with resource competition (**??**). In these approaches the relative orders between species remain fixed, but other models allow species relative orders and timings to vary year-to-year assuming the underlying relationship between pairwise differences in timings and interspecific-competition is known—and static. Each of these approaches in turn thus fixes one aspect of environment $\times$ phenology $\times$ competition relationship, an understandable simplification, but one that may make forecasting community shifts with climate change challenging.

Community assembly models including phenology have all included environmental variability (**??**) assuming an underlying stationary environment (one that varies year-to-year but with an underlying constant distribution), but climate change represents a clearly non-stationary environment. Temperature, for example, trends higher in its mean over time, with potential additional changes to the variance (**?**). Because environmental variability in assembly models structures what species combinations are stable, shifts in the environment with anthropogenic climate change are likely to reshape communities, but forecasting them requires accurately capturing how the environment affects phenology and competition together in the simplest—but still robust—form. We argue this means capturing how dynamically the environment can determine both absolute and relative timings of species, and the how those timings directly impact co-existence. While current modeling approaches to date have simplified one aspect of this problem, it can be tractably addressed through new models that explicitly allow these dynamics in both inter- and inter-annual time, and connect to the underlying physiology of phenology.

4 Climate Change, Phenological Sensitivity, Realized Phenology and Coexistence

4.1 Climate change alters relationship between phenogical sensitivityand realized phenology differences

First to understand the effects of climate change on realized phenological differences between competing species, we ran our model under two climate change scenarios; one with a mean winter chilling period of 12 weeks and one with a winter chilling period of 6 weeks (Figure 2a). With 12 weeks of winter chilling, the differences in realized germination phenology of compete species was minimized, with realized phenological differences being less than 2 days in approximately 50% of model runs (Figure 2b). Exposure to only an average 6 week of winter chilling increased the realized phenological differences between competing species with only about 10% of of model runs producing differences in realized germination phenology of less than 2 days (Figure 2b).

The phenological sensitives of species were drawn from the same distribution under each scenario, so the shift in realized phenological differences must be a product of the interaction between cliamte and phenological sensitivty. In fact, we see that the relationship between phenolgical sensitivity differences and realized phenological differences shifts in assoicated with the changed climate, with the slope of the relationship becoming steeper under 6 weeks of chilling (put in parameters here; Figure 2c). Some transition statement or phrase. Next we must consider how this shift affects the tradeoff between phenology and competition.

4.2 Climate change alters the tradeoff between phenological sensitivity and competition

Because of the strong dependency on the environment and the observed shift in the relationship between phenological sensitivity and realized phenology under each climate scenario, we would expect that this shifting enviroment would alter aspects of the trade off between phenology and competition, but exactly how is not neccisarily straight forward.

First, we find the fundamental tradeoff between realize phenological differences and competitive differences does not change in the new environment (Figure 3a). This is not surprising, since the degree that phenological can offset competitive differences in fixed (say better). However, while in our models we treat inherant competition differences as fixed, it is important to note that the shift in environment changes the phenological differences of virtually all species pairs in our simulation, which means those that coexisted under one set of environmental conditions cannot coexist in a new environment, as highlighted in the circled points in Figure 3a. So while the fundamental relationship between realized phenology and competitive ability doesn't change with climate change, the species that can coexist do (say better).

Because phenological sensitivity is the mediator between environmental variation and realized phenology, the tradeoff between sensitivity differences and competitive difference must shift in accordence with the environment to maintain the realized phenolgical differences that produce coexistence. In fact we see that under 6 weeks of chilling, the slope of this relationship become steeper (put in slope value here, Figure 3b), because larger differences in competitive ability are necessary to maintain the trade off in Figure 3a.

Paragraph discussing ecological implications? eg. While our simulations are highly simplified version of a real ecological community, we can make some predictions about the kinds of species that will coexist under a novel climate. More phylogenetically distinct. Different life hystory, dormancy types. Maybe this is too hand wavy.

5 Paths Foward

Predicting assembly in an era of global change will require the closer integration of experiments and theory (?), and enhancing the biological realism of both. Here, we demonstrate a simple way incorporating a more biological realistic germination frame into assembly models that produced important insights about how the relation between phenological sensitivity and competitive ability will shift in order the maintain the relationship between realized phenological difference and competitive differences (Figure Figure 3). However there are several other import aspects (something) that could be integrated into community assembly experiments and models to increase their relevance to on-the ground biology (say better).

1. Phenology is not the only thing that varies with the environment trait and density dependent mechanisms (rudoff paper)/ ie, we vary germination fraction and phenology. (note that our model can do this)
2. multiple interacting cues
3. add risk of being early
4. other life histories, i.e, perennials, trees.

Figures

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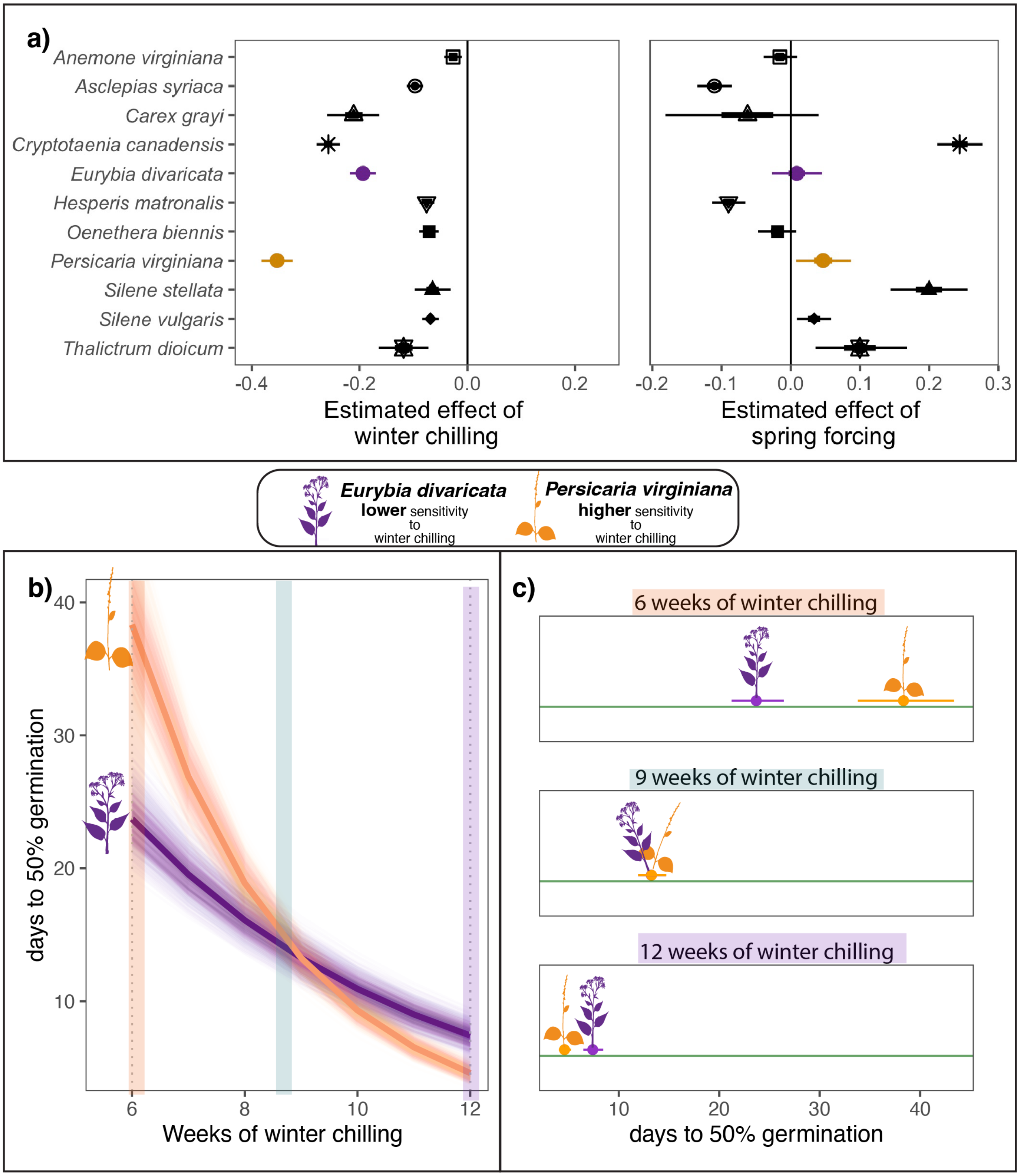

Figure 1: **Co-occur species can differ substantially in their phenological sensitivity (Δ phenological event/Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals. In b) and c) we highlight the phenological sensitivity to winter chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrations how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions. Point represent the predicted mean number of days to 50% germination and bars the 95% uncertainty intervals

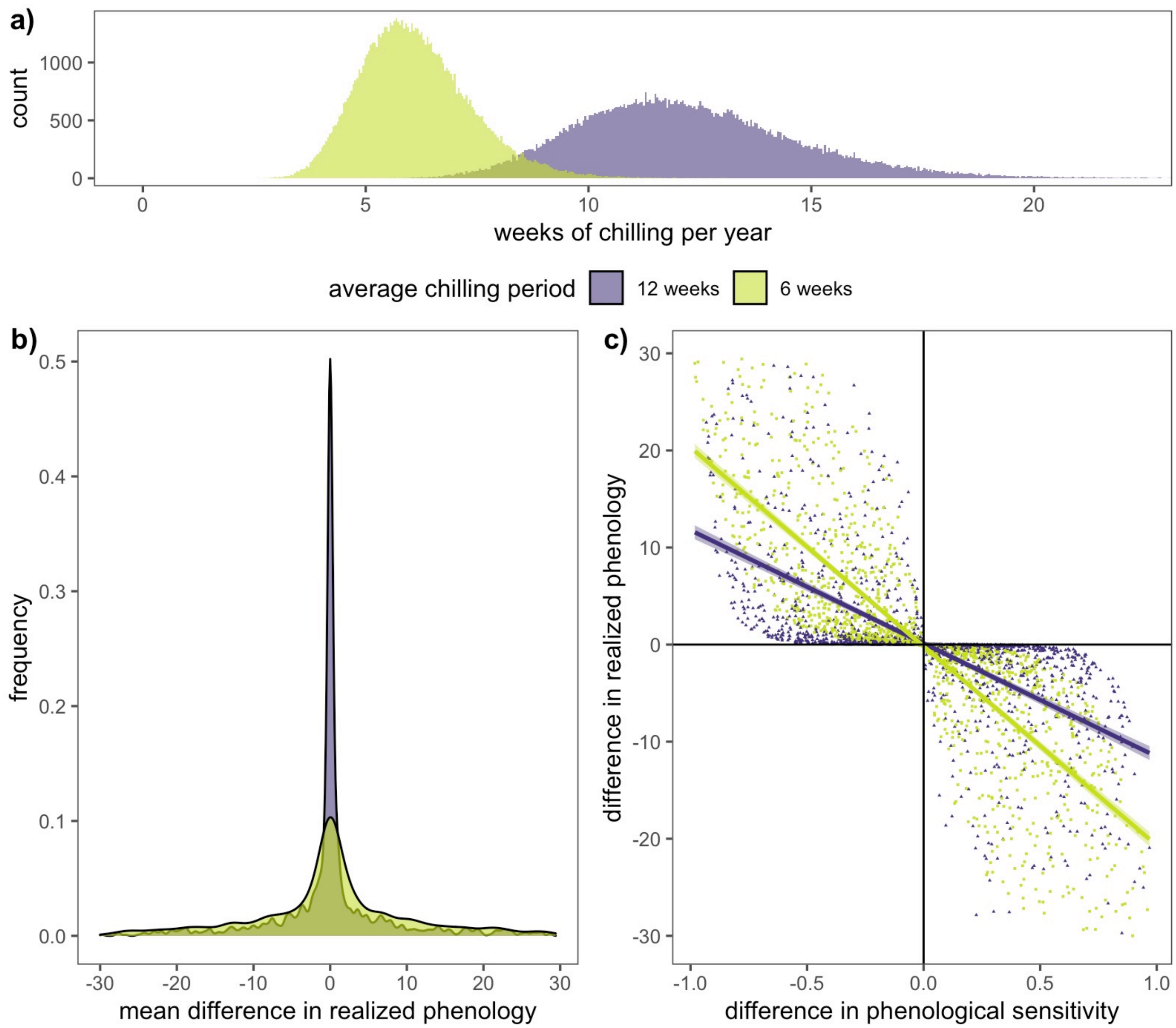

Figure 2: **Climate change will the relationship phenological sensitivity differences and realized phenological differences.** Panel a) shows how climate change will shift the distribution of the duration of chilling ecological communities receive each winter. Panel b) shows the frequency distributions for the average difference in realized germination phenology under both climate scenarios—shorter chilling durations result in larger realized phenological differences between species. These changes occur because the relationship between phenological sensitivity differences and realized phenology differences, depicted in panel c), are dependent on the environment, and a change in chilling conditions alters this relationship.

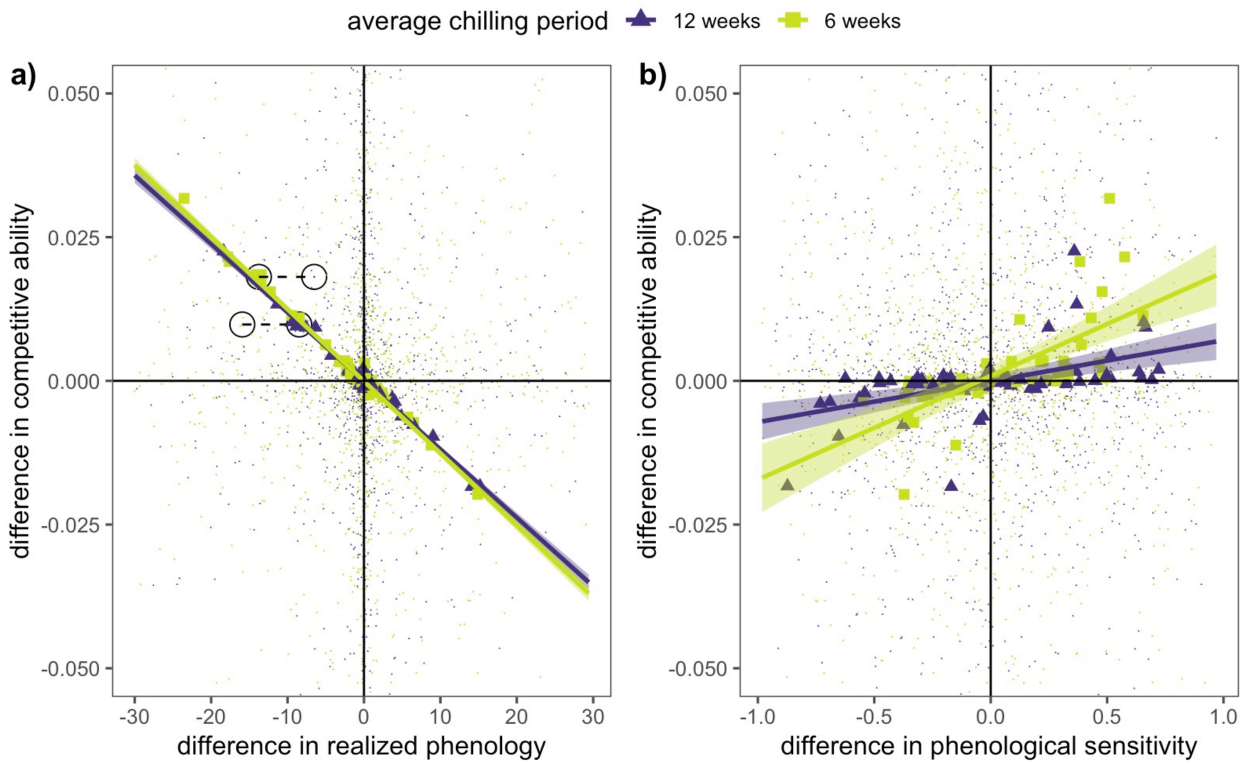

Figure 3: **Climate change will alter the relationship competititive ability, phenological sensitivty, realized phenology and coexistence.** Panel a) depicts the trade-off between realize phenological differences and competitive differences. While the tradeoff relationship between realized phenology and competition does not shift with climate change, species that coexisted under one climate scenario will not coexist if climate conditions shift, as highlighted by the two sets species pairs circled in panel a). Panel b) depicts the trade-off between phenological sensitivity differences and competitive differences under two climate scenarios (average of 12 and 6 weeks of chilling). With climate change, the trade-off between phenological sensitivity and competition does shift because less chilling amplifies the realized phenological differences among species